ADITI SABNE

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PROFESSIONAL SUMMARY

Result Driven Software Engineer with a strong background in Software Development, AI/ML and Database Management. Delivered 3+ end-to-end projects using Python, Java, MySQL, Flutter, Firebase, and OpenCV with hands-on experience in REST APIs, database-driven applications, and computer vision systems. Built practical solutions involving Real-time Features, Data-driven Decision-making, and Spearheaded development of SECURA women's safety app integrating native Android SMS alerts via MethodChannel, cutting emergency response latency by 20%. Demonstrated technical capabilities through a National-Level Smart Cops Hackathon 2025 Win, competing against 1,322 registered participants.

EDUCATION

B.Tech - Electronics and Computer Science — PVPIT, Budhgaon , Sangli (DBATU University) 2026
HSC (XII) — Nav Krishna Valley School, Sangli 2022
SSC (X) — Alphonsa School, Miraj 2020

TECHNICAL SKILLS

- **Programming**: Python | Java | C/C++ | HTML
- **Software Development**: Flutter | OOP |REST APIs | JSON | API Integration
- **Backend & Database**: MySQL | SQL | PostgreSQL | Firebase | Firestore | DBMS
- **Data Structures & Algorithms**: Arrays | Strings | Linked Lists | Stacks | Queues | Hash Tables | Trees | Graphs | Heaps | Recursion | Sorting | Searching |Time & Space Complexity
- **AI / Machine Learning**: Machine Learning |Natural Language Processing | Computer Vision
- **DevOps & Cloud**: CI/CD | GitHub Actions | Linux
- **Tools**: Git | GitHub | VS Code | Android Studio
- **Core CS Fundamentals**: Data Structures & Algorithms | DBMS | Operating Systems | Computer Networks | Computer Architecture | OOP

PROJECTS

SECURA — Women's Safety Companion App | *Flutter, Dart, Firebase, Gemini API* **Demo**

- Forged native Android integration via MethodChannel, enabling critical SMS alerts and improving user safety response times by 20%.
- Integrated Firebase Authentication and Firestore, enabling seamless user management and real-time data operations; **drove 25% increase** in user engagement metrics.
- Debugged cross-platform integration issues across Flutter, native Android, and third-party APIs; **won 1st place at Smart Cops Hackathon 2025**.; recognized by **Sangli Cyber Police.**

SQL Query Engine & Database Management System / *C++, DBMS, DSA, Data Structures, Algorithms* **Demo**

- **Developed a custom SQL query engine in C++**, implementing query parsing and execution for core operations including SELECT, INSERT, UPDATE, and DELETE, with persistent data storage.
- **Implemented B+ Tree and hash-based indexing** to optimize data retrieval, with query-performance analysis comparing indexed search against linear search using execution time and comparison metrics.
- **Engineered core DBMS components** including query processing, storage management, transactions, concurrency control, and error handling, applying DSA and DBMS concepts to build a functional database engine.

PresentX – Student & Teacher Connector | *Flutter, Firebase, Supabase* **Demo**

- Developed a cloud-based academic management platform supporting attendance, assignments, role-based authentication, and real-time synchronization.
- Implemented secure backend services using Firebase, Supabase, REST APIs, and cloud database technologies.

ACHIEVEMENTS & CERTIFICATIONS

- **Winner — Smart Cops Hackathon 2025** (National-Level), Honoured by Sangli Cyber Police for developing an AI-powered women's safety application, by developing an AI-powered women safety application.
- **Second Prize — Nirmiti 2025, Electro-Mania,** awarded for excellence in C Programming, General Aptitude and Digital Logic.
- Completed **Industry 4.0 and IOT** workshop conducted by **triMAP Tech Solutions**